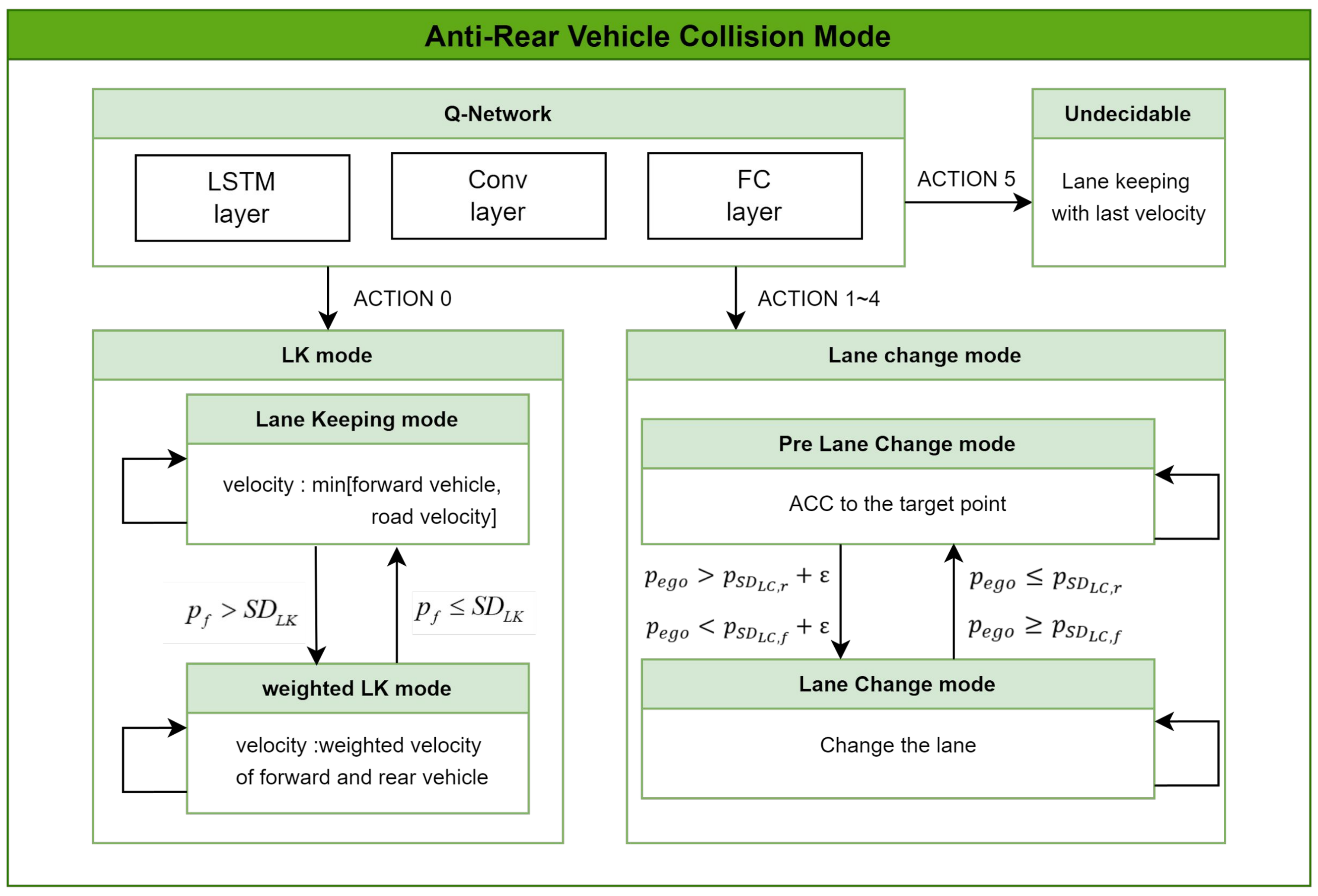
Development of a Rear Approach Vehicle Collision Prevention Algorithm using Deep Reinforcement Learning

Jewoo Kwak, Yonghwan Jeong, Xianjun Li, Taeyeon Lee, DongHyun Kim,
Department of Automotive Engineering,
Seoul National University of Science and Technology, Korea
e-mail : 23512093@seoultech.ac.kr

1. Introduction

- Fatalities from accidents involving commercial vehicles such as trucks and buses are the highest among vehicle-to-vehicle collisions.
- Proactive avoidance algorithms for inattentive rear vehicles, such as those caused by drowsy driving or lack of attention, could reduce the number of fatalities in rear-end collisions.
- In this study, we used DQN, which is known for its effectiveness with discrete input and output states, to find the optimal empty space to avoid collisions with unconscious rear vehicles.
- We defined the input state as the 8 surrounding vehicles and the output action as 6 modes, covering longitudinal and lateral control.

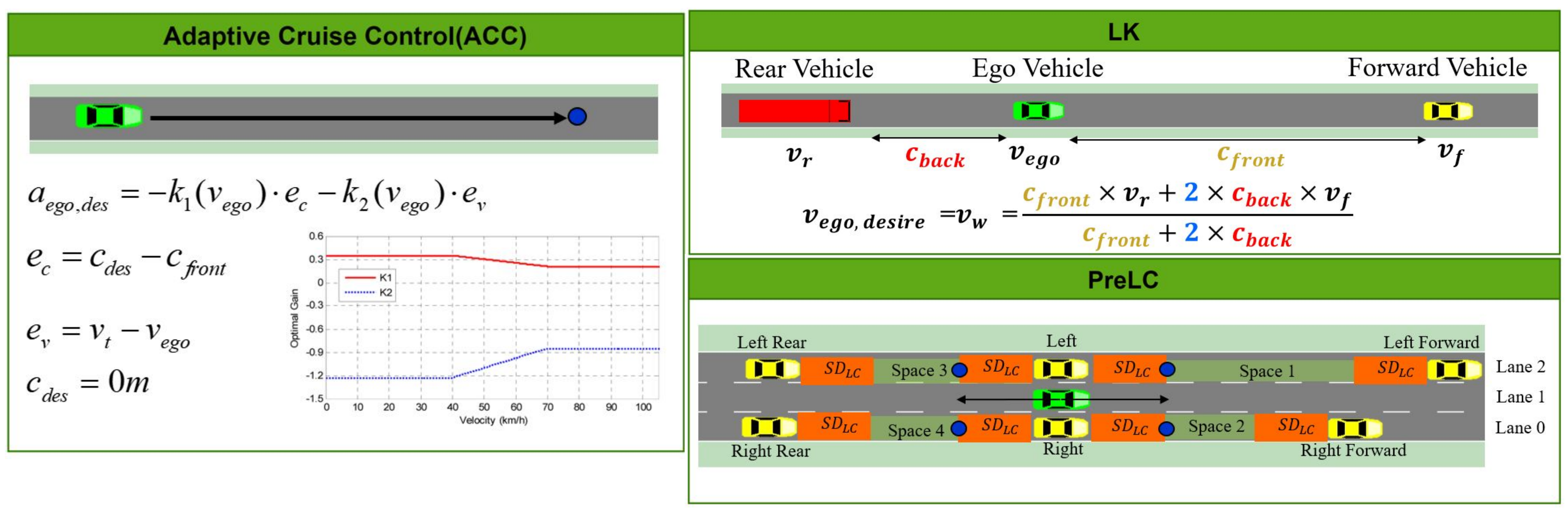
2. Architecture



- The Q-network from DQN determines the driving mode.
- The mode change condition is based on a safety distance derived from statistics of human driving data.

3. Control logic

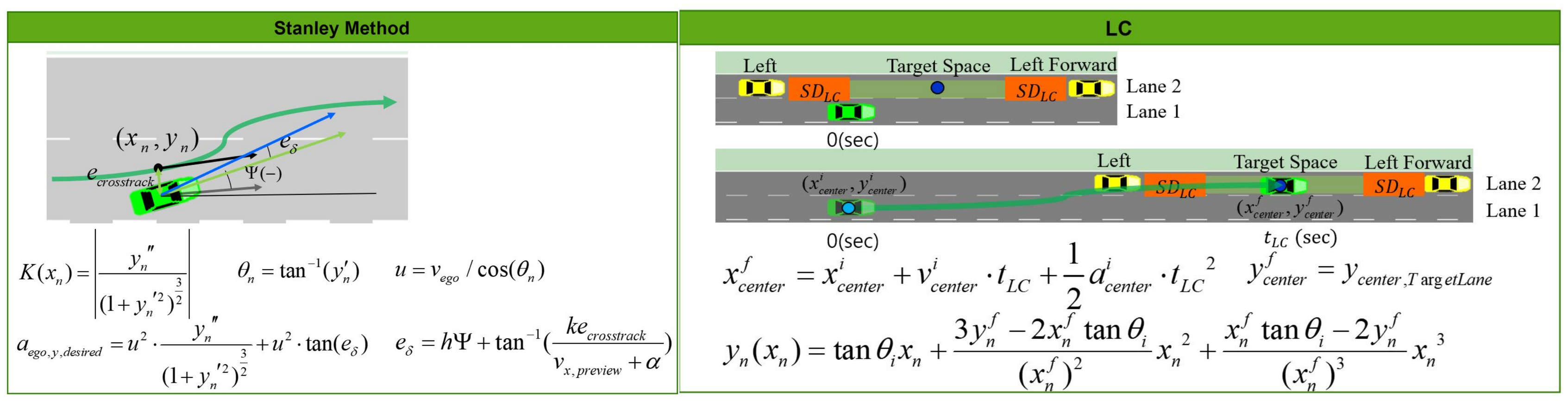
Longitudinal Control



$$a_{ego,des} = -k_1(v_{ego}) \cdot e_c - k_2(v_{ego}) \cdot e_v$$
$$e_c = c_{des} - c_{front}$$
$$e_v = v_t - v_{ego}$$
$$c_{des} = 0m$$
$$v_{ego,desire} = v_w = \frac{c_{front} \times v_r + 2 \times c_{back} \times v_f}{c_{front} + 2 \times c_{back}}$$

- ACC controls velocity error and the clearance error between the target point and the ego vehicle.

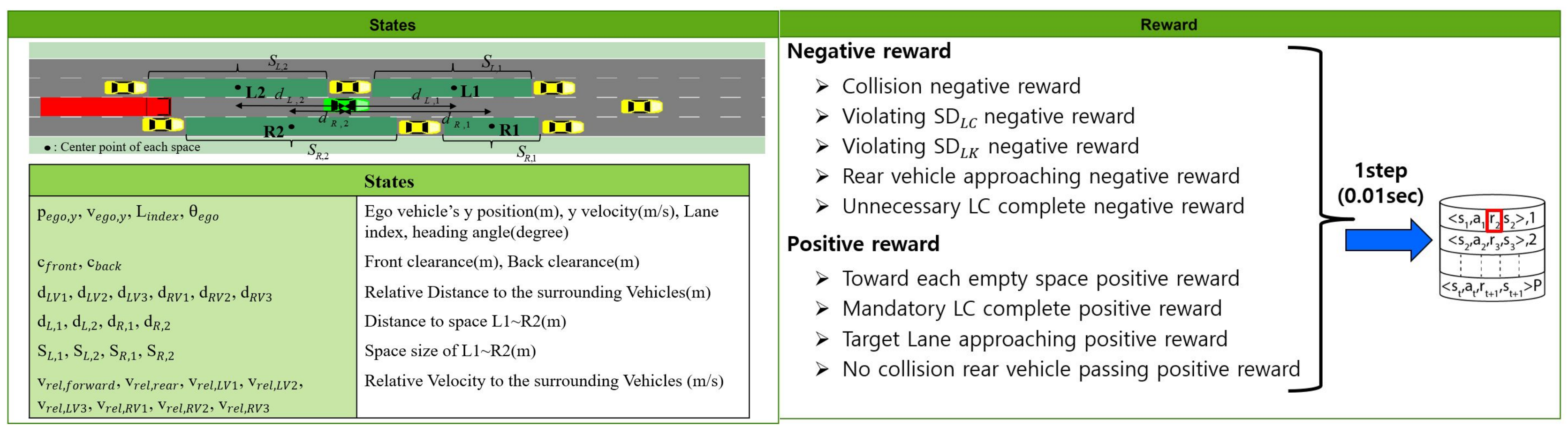
Lateral Control



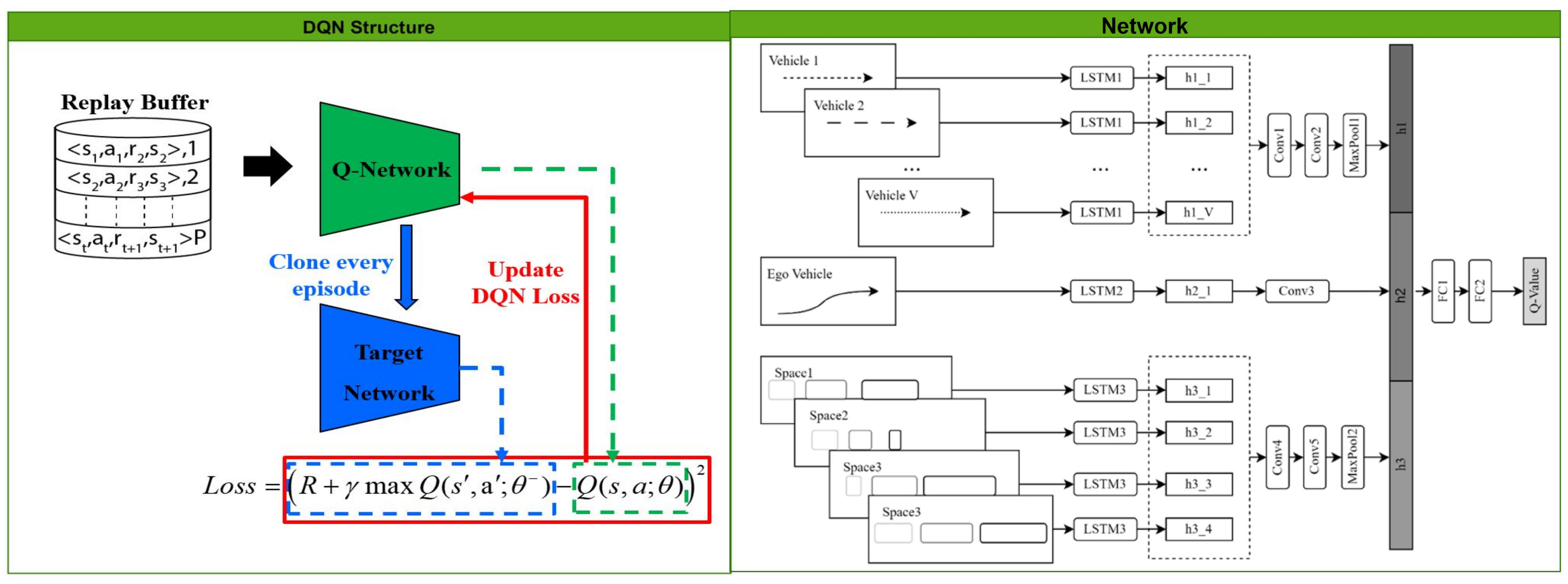
$$K(x_n) = \frac{y_n''}{|(1+y_n'^2)^{\frac{3}{2}}|} \quad \theta_n = \tan^{-1}(y_n') \quad u = v_{ego}/\cos(\theta_n)$$
$$a_{ego,y,desired} = u^2 \cdot \frac{y_n''}{(1+y_n'^2)^{\frac{3}{2}}} + u^2 \cdot \tan(e_\theta) \quad e_\delta = h\Psi + \tan^{-1}\left(\frac{ke_{crosstrack}}{v_{x,preview} + \alpha}\right)$$
$$x_{center}^f = x_{center}^i + v_{center}^i \cdot t_{LC} + \frac{1}{2} a_{center}^i \cdot t_{LC}^2 \quad y_{center}^f = y_{center,TargetLane}$$
$$y_n(x_n) = \tan\theta_i x_n + \frac{3y_n^f - 2x_n^f \tan\theta_i}{(x_n^f)^2} x_n^2 + \frac{x_n^f \tan\theta_i - 2y_n^f}{(x_n^f)^3} x_n^3$$

- The Stanley method is applied to a cubic polynomial curve to compensate for heading and cross-track errors.

4. DQN



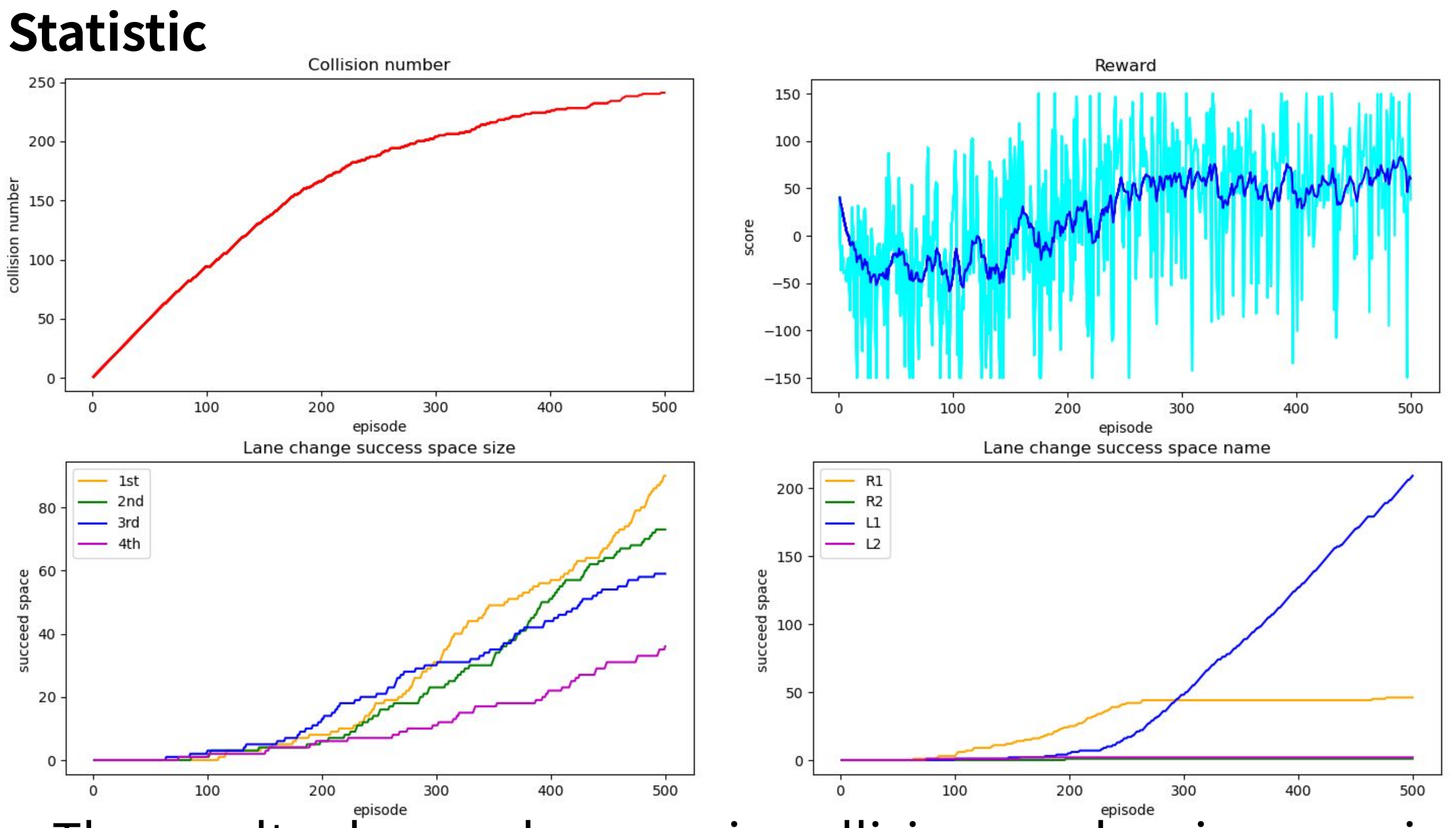
- The state-action-reward-next state pairs are collected in a replay buffer, from which the Q-network learns.



- DQN uses two networks: the Q Network and the Target Network, both with LSTM, convolutional, and fully connected layers.
- The Q Network outputs actions based on the current state at each step, and the Target Network is cloned from it every episode. The Q Network learns by minimizing the mean squared error (MSE) loss between the networks.

5. Analysis Results

Statistic



- The results show a decrease in collisions and an increase in successful lane changes to the largest available space.

4. Conclusion

- In this study, we define the state, action, and reward components for Deep Q-Network (DQN) to determine the optimal empty space for proactively avoiding inattentive vehicles approaching from behind.
- In future research, we will compare the performance of advanced deep reinforcement learning techniques, including A3C, DDPG, and SAC.

Autonomous & Intelligent Mobility Laboratory